MATH 1342 – Elementary Statistics – Week 5: Two-Variable Relationships and Linear Regression

Sep 21 – 23 | 2.4 Two Variable Relationships • 2.5 Two Quantitative Variables: Linear Regression | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CLO 2) **2.** Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r. (CLO 7) **3.** Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Participation: Box Plots and Outliers – Mon: 1.5×IQR rule review; side-by-side boxplots (tea vs. coffee)	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 27, 11:59 PM)	WileyPLUS Blackboard
4. Interpret the slope and intercept of a regression line in context with correct units, and judge whether the intercept is meaningful. (CLO 7) **5.** Use a regression equation to predict a response value, and compute and interpret a residual. (CLO 7) **6.** Use technology to find the least-squares regression line and explain what “least squares” means. (CLO 7) **7.** Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, influential outliers, and causal claims. (CLO 7)	e-Text: 2.5 Two Quantitative Variables: Linear Regression Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) Slides: 2.5 Two Quantitative Variables: Linear Regression	Examples: baseball game length and commute-time regression lines (predict, interpret slope, find a residual)	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)		Participation: Two Variable Relationships – Wed: NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier; Your Turn: spotting residuals (MO2, MO3, MO5)		
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Practice Test 1 (Blackboard, due Thu Oct 1, 11:59 PM; 120 min, proctoring practice)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 2 Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 7** Solve linear regression and correlation problems.